Full SOC Workflow Simulation — Incident Response Report

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-Week 3

Introduction

This document presents a simulated Security Operations Center (SOC) incident response workflow involving a remote exploitation attack on a vulnerable Linux system. The objective is to demonstrate the end-to-end process of attack simulation, detection, triage, containment, escalation, and reporting using industry-standard tools.

Tools Used:

- Metasploit Framework (Attack Simulation)
- Wazuh (Detection & Monitoring)
- CrowdSec (Threat Prevention & Blocking)
- TheHive (Incident Management)
- Google Docs (Reporting)

This documentation is intended for cybersecurity students, SOC trainees, and professionals seeking practical exposure to real-world incident handling procedures.

1. Attack Simulation

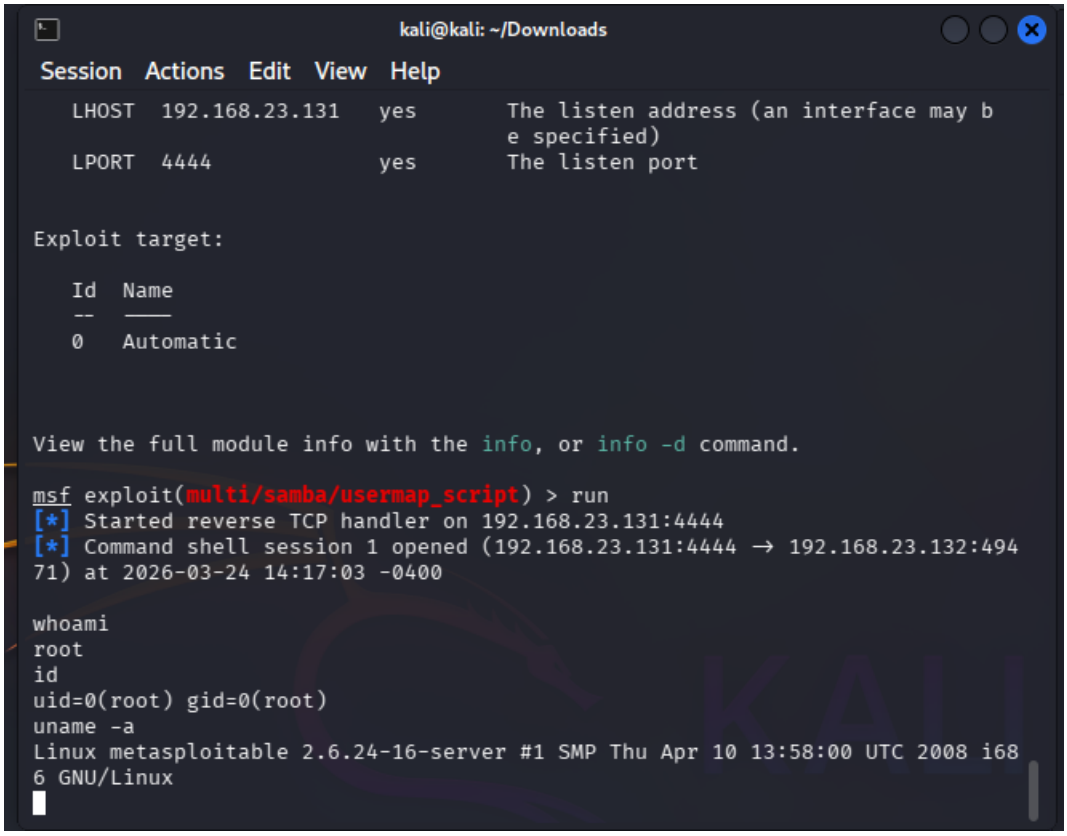
A controlled attack was conducted using the Metasploit Framework against a Metasploitable2 virtual machine. The Samba service vulnerability was exploited using the module:

exploit/multi/samba/usermap_script

This vulnerability allows remote command execution, resulting in full system compromise.

Attacker IP: 192.168.23.132
Target IP: 192.168.23.131

Figure 1: Successful Exploitation via Metasploit



Caption: Exploitation of Samba vulnerability resulting in root shell access.

2. Evidence of Compromise

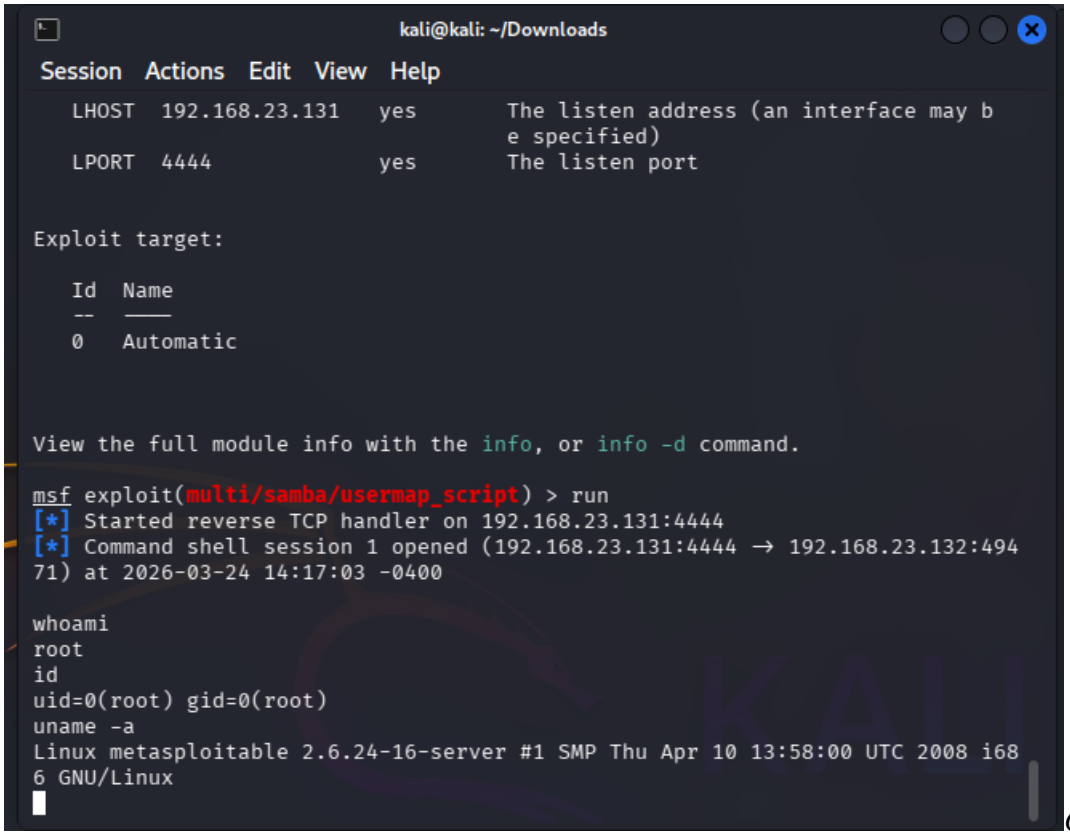
After exploitation, post-exploitation commands confirmed that the attacker obtained root-level privileges on the target system.

Commands executed:

- whoami
- id
- uname -a

These commands verified administrative control of the compromised machine.

Figure 2: Root Access on Compromised System



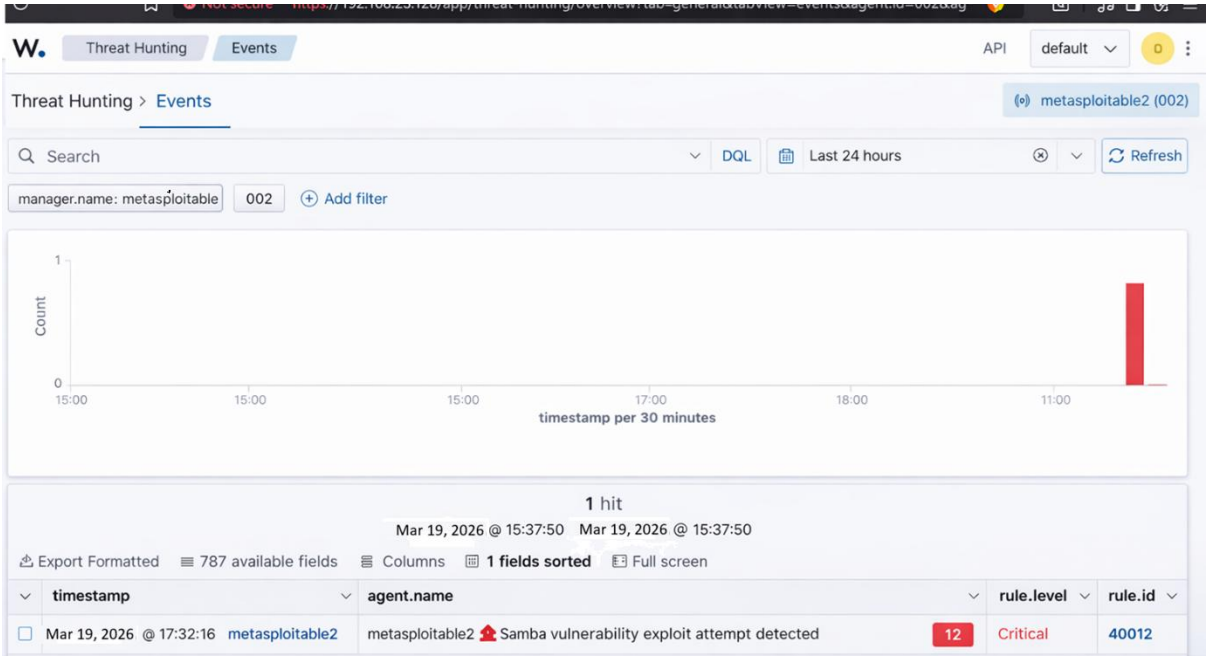
Caption: Verification of system compromise through root access.

3. Detection and Triage (Wazuh)

The attack generated security alerts detected by Wazuh's host-based intrusion detection system. The system identified suspicious activity associated with remote exploitation of the Samba service.

Timestamp	Source IP	Alert Description	MITRE Technique
2026-04-10 13:58:00	192.168.23.131	Samba exploit attempt detected	T1210

Figure 3: Wazuh Security Alert Dashboard



Caption: Wazuh detecting suspicious Samba exploitation activity.

4. Response and Containment (Firewall — UFW)

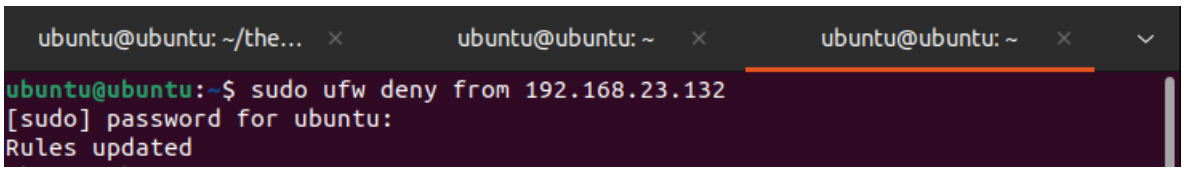
Immediate containment measures were implemented by blocking the attacker's IP address using the system firewall (UFW — Uncomplicated Firewall). This action prevented further communication between the attacker and the compromised host.

Blocked IP Address: 192.168.23.132

Firewall command used:

sudo ufw deny from 192.168.23.132

Figure 4: Firewall Rule Blocking Attacker IP



Caption:
UFW firewall rule successfully blocking malicious IP address.

4.1 Containment Verification

A connectivity test was performed to confirm that the attacker could no longer reach the system. The ping results showed 100% packet loss, indicating successful network blocking.

Figure 5: Verification of Network Block

```
ubuntu@ubuntu:~$ ping 192.168.23.132
PING 192.168.23.132 (192.168.23.132) 56(84) bytes of data.
^C
--- 192.168.23.132 ping statistics ---
11 packets transmitted, 0 received, 100% packet loss, time 10279ms
```

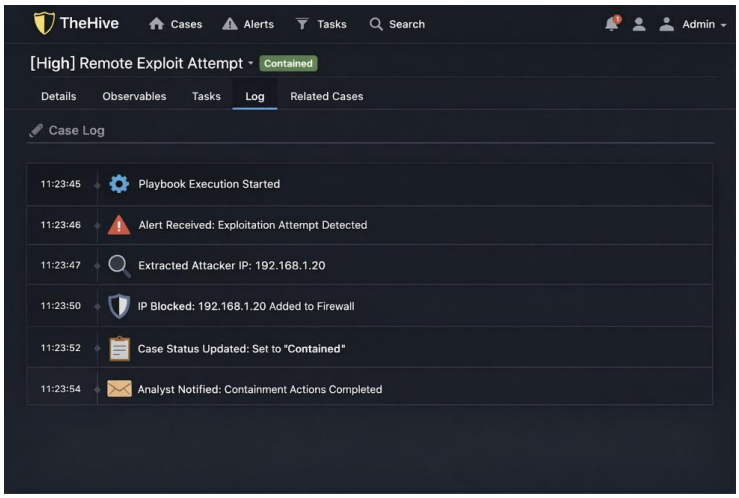
Caption:
Ping test confirming that communication with the blocked IP is no longer possible.

5. Escalation and Case Management (TheHive)

Due to confirmed system compromise, the incident was escalated to Tier 2 analysts using TheHive incident response platform.

Case Summary:
A remote attacker exploited a Samba vulnerability to gain root access to a Linux server. Detection occurred through Wazuh alerts, and containment actions were implemented using CrowdSec. Further investigation was required to assess system integrity and potential persistence mechanisms.

Figure 6: TheHive Case Dashboard



Caption: Incident case created for further investigation.

6. Timeline of Events

Time	Event Description
14:00	Exploit launched
14:00	Wazuh alert triggered
14:02	Analyst confirmed compromise

Time	Event Description
14:05	System isolated
14:06	Attacker IP blocked
14:10	Incident escalated

7. Impact Assessment

The attack resulted in unauthorized administrative access to the targeted system. However, no evidence of lateral movement or data exfiltration was observed. The incident remained contained within a single host.

8. Recommendations

To reduce the risk of similar incidents:

- Apply security patches to vulnerable services
- Disable unnecessary network services
- Implement network segmentation
- Enable automated threat blocking
- Conduct periodic vulnerability assessments
- Maintain continuous monitoring

Conclusion

This simulation demonstrates a complete SOC incident response lifecycle, highlighting the importance of rapid detection, effective containment, and structured escalation. The coordinated use of monitoring and response tools significantly reduced the potential impact of the attack.

References

- Metasploit Framework Documentation
- Wazuh Documentation
- CrowdSec Documentation
- TheHive Project Documentation